AGRICOLA

SMART FARMING PROJECT • CONNECTION GUIDE

1. Sensor & OLED Connections

Component	Module pin	Arduino pin
Soil moisture sensor	Analog OUT	**A0**
Heartbeat sensor	Signal OUT	**A1**
Touch sensor	OUT	**D2**
OLED display	SDA	**A4 (SDA)**
OLED display	SCL / SCK	**A5 (SCL)**

2. L298N Motor Driver Connections

Component	Driver / signal	Connect to
L298N motor driver	ENA	**D6**
L298N motor driver	IN1	**D7**
L298N motor driver	IN2	**D8**
Water pump	Pump wires	**OUT1 / OUT2**
External pump supply	Positive	**L298N motor +**

3. Power & Ground

Component	Pin	Connection
Soil sensor / Touch sensor	VCC (check module)	**Arduino 5V**
Heartbeat sensor / OLED	VCC	**Use module-rated voltage**
All sensor modules	GND	**Arduino GND**
L298N + pump supply	GND / negative	**Common ground**

IMPORTANT SAFETY NOTES

- Do not power the water pump directly from Arduino 5V.
- Use a suitable external supply for the pump and L298N.
- Connect Arduino GND and driver/supply negative together.
- Confirm your sensor and OLED voltage ratings before wiring.
- This table follows the pin assignments in the supplied sketch.